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AIM : Data Wrangling with pandas: Manipulate and transform data to suit the analysis needs.

1 . What is data wrangling? Difference between data wrangling and data preprocessing?

Data wrangling is the process of cleaning, transforming, organizing and structuring raw data into a suitable format for analysis. It involves handling missing values, correcting inconsistencies, filtering unwanted data and converting data types. Data preprocessing is a broader step mainly used before applying machine learning models and includes tasks such as normalization, encoding categorical values and scaling features. In simple terms, data wrangling prepares raw data for analysis, while data preprocessing prepares cleaned data for modeling and prediction tasks. 2 .

Importance of Exploratory Data Analysis (EDA) in Pandas?

ANS:

Exploratory Data Analysis (EDA) is important because it helps understand the structure, quality and patterns present in the dataset before performing detailed analysis or building models. Using Pandas, EDA allows users to inspect datasets, summarize statistics, detect missing values, identify outliers and understand relationships between variables. This process improves decision making and ensures the dataset is suitable for further analysis, which increases the accuracy of results.

3 . Differences between Series, DataFrame and Panel?

ANS:

A Series is a one-dimensional labeled data structure that stores a single column of data with an index. A DataFrame is a two-dimensional data structure that stores data in rows and columns like a table and is the most commonly used structure in Pandas. A Panel is a three-dimensional data structure used to store multiple DataFrames together, but it is now deprecated in modern Pandas versions and replaced by other tools like multi-index DataFrames. These structures help organize data based on complexity and analysis requirements.

4 . Use of isnull(), notnull(), dropna(), fillna()?

ANS:

The `isnull()` function is used to detect missing values in a dataset by identifying

5. What is column transformation and why is it important in data analysis?

ANS:

Column transformation is the process of modifying or converting column values to

```
import pandas as pd
import numpy as np

data = {
    "Name": ["Amit", "Aditya", "Uday", "krishana", None],
    "Age": [20, 21, None, 19, 22],
    "Marks": [85, 90, 78, None, 88],
    "City": ["Pune", "Mumbai", "Pune", "Russia", "Mumbai"]
}

df = pd.DataFrame(data)
print("Original Data:\n", df)

# Check missing values
print("\nMissing Values:\n", df.isnull())

# Fill missing values
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Marks"] = df["Marks"].fillna(0)
df["Name"] = df["Name"].fillna("Unknown")

# Drop rows with any remaining null values
df = df.dropna()

# Change data type
df["Age"] = df["Age"].astype(int)

# Rename columns
df.rename(columns={"Marks": "Score"}, inplace=True)

# Add new column (Transformation)
df["Result"] = df["Score"].apply(lambda x: "Pass" if x >= 40 else "Fail")

# Filter data
```

```

filtered_df = df[df["Score"] > 80]

# Sort data
sorted_df = df.sort_values(by="Score", ascending=False)

# Grouping data
grouped = df.groupby("City")["Score"].mean()

# Display results
print("\nCleaned Data:\n", df)
print("\nFiltered Data (Score > 80):\n", filtered_df)
print("\nSorted Data:\n", sorted_df)
print("\nGrouped Data (Average Score by City):\n", grouped)

```

Original Data:

	Name	Age	Marks	City
0	Amit	20.0	85.0	Pune
1	Aditya	21.0	90.0	Mumbai
2	Uday	NaN	78.0	Pune
3	krishana	19.0	NaN	Russia
4	None	22.0	88.0	Mumbai

Missing Values:

	Name	Age	Marks	City
0	False	False	False	False
1	False	False	False	False
2	False	True	False	False
3	False	False	True	False
4	True	False	False	False

Cleaned Data:

	Name	Age	Score	City	Result
0	Amit	20	85.0	Pune	Pass
1	Aditya	21	90.0	Mumbai	Pass
2	Uday	20	78.0	Pune	Pass
3	krishana	19	0.0	Russia	Fail
4	Unknown	22	88.0	Mumbai	Pass

Filtered Data (Score > 80):

	Name	Age	Score	City	Result
0	Amit	20	85.0	Pune	Pass
1	Aditya	21	90.0	Mumbai	Pass
4	Unknown	22	88.0	Mumbai	Pass

Sorted Data:

	Name	Age	Score	City	Result
1	Aditya	21	90.0	Mumbai	Pass
4	Unknown	22	88.0	Mumbai	Pass
0	Amit	20	85.0	Pune	Pass
2	Uday	20	78.0	Pune	Pass
3	krishana	19	0.0	Russia	Fail

Grouped Data (Average Score by City):

City	
Mumbai	89.0
Pune	81.5
Russia	0.0

Name: Score, dtype: float64

Conclusion :

Data wrangling with Pandas is an important step in preparing raw data for analysis by cleaning, organizing and transforming it into a meaningful format. It helps handle missing values, detect inconsistencies and improve the overall quality of the dataset. Exploratory Data Analysis supports better understanding of data structure and patterns before applying further techniques. Structures like Series and DataFrame allow efficient storage and manipulation of data, while functions such as `isnull()`, `notnull()`, `dropna()` and `fillna()` help manage missing values effectively. Column transformation further improves data usability by converting it into a suitable format for analysis, leading to more accurate and reliable results in data science tasks.